

Sheng-Kai (Kyle) Wen

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Experience

PuckAI — Live AI Platform

May 2025 - Present

AI Engineer — Independent (via UMass CICS CICStep)

thepuckai.com

- **Build a production Retrieval-Augmented Generation (RAG) chatbot over 7,643** grounded reports, with **hybrid dense (pgvector HNSW) and lexical retrieval** fused by Reciprocal Rank Fusion, **Voyage AI** embeddings, and reranking.
- **Take scope-limited query relevance from NDCG@10 0.38/0.31 to 1.00** on a **77-query benchmark** by **engineering a deterministic alias-index router** feeding a SQL pre-filter, after diagnosing that embeddings cannot encode current team membership.
- **Ship 7,643 grounded reports — 7,643 validated, 0 quarantined — by constraining the LLM to narrate only pipeline-computed numbers** behind a deterministic validator (numeric trace, proper-noun trace, banned vocabulary) run in a mandatory self-check loop, **raising first-pass validation from 0% to 100%**.
- **Build a retrieval evaluation harness (NDCG@10, MRR, Recall@K, P@K)** run as a benchmark suite against a labeled query set, catching multiple silent retrieval-correctness regressions before they shipped.
- **Tune fusion to a 2:1 dense-to-lexical ratio** after measuring equal weighting underperformed *both* legs alone, and **keep model choice reversible** behind a capability-gated multi-provider layer (Claude, Gemini, OpenAI).

Honda Research Institute (HRI) & 99P Labs

Feb 2026 - May 2026

Graduate Research Extern — Hindsight team (Advised by Prof. Andrew McCallum)

- **Architected Hindsight, a knowledge-graph retrieval layer** fusing **BM25** with agentic, **intent-conditioned graph traversal** — an LLM labels each question leading, lagging, or exploratory to steer edge direction — over open models (Qwen3-8B/32B) with **AST-based indexing**.
- **Ran quantitative experiments for evidence-based decisions**, lifting file-level localization to **Acc@1 0.535** (vs 0.454 BM25; +12.9 pts with no location hint) through controlled ablations.
- **Designed evaluation infrastructure and metrics** — Fact Recall, Acc@k, and **LLM-judged answer quality** — across **DeepCodeBench** (1,144 QA pairs) and **LocBench** (560 tasks); **co-authored the 14-page report**.
- **Traced failing tests to the source locations responsible by engineering a LocAgent-based test-linkage layer**, grounding agent reasoning in verifiable evidence rather than similarity.

Fan2see, Taipei, Taiwan

Apr 2021 - Apr 2022

Co-Founder & Full-Stack Engineer

- **Shipped real systems under load by building a full-stack PERN product** (React, Node.js, PostgreSQL) to hundreds of active users with end-to-end ownership, driving **30% month-over-month growth**.
- **Reduced media-processing latency 25% and improved scalability 40%** by **architecting an asynchronous AWS Lambda/S3 pipeline** for high-volume media workloads.

Applied AI Projects

MAIN-RAG — Multi-Agent Filtering for RAG (UMass Information Retrieval)

Sep 2025 - Dec 2025

- **Reimplemented the Predictor-Judge-Generator pipeline from scratch** with a judge scoring each (query, document, answer) tuple by **logit margin**, then **replaced the expensive LLM judge with rule-based thresholds** (Otsu's method, quantile, mean/std) matching adaptive-threshold quality across **four datasets**; **first author**.

Timestamp-Aligned Summarization from ASR Transcripts (UMass Advanced NLP)

Sep 2025 - Dec 2025

- **Reached ROUGE-L 59.83 with BERTScore F1 91.44 at 80.43% segment-alignment accuracy** on MeetingBank by **fine-tuning BART** for segment titles with a **BERT-based self-retrieval aligner**.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, Java, C/C++

Generative AI & LLM: Retrieval-Augmented Generation (RAG), prompt engineering, grounded generation & deterministic output validation, LLM integration (Claude, GPT-4o, Gemini, Qwen3, Llama-3.1), AI agents & agentic workflows, fine-tuning (BART/BERT), batch pipelines, LLM-as-judge, answer caching, multi-provider routing

Retrieval & Evaluation: vector search (pgvector, HNSW, FAISS), embeddings & reranking (Voyage AI, Sentence-BERT), hybrid dense + lexical retrieval, Reciprocal Rank Fusion, BM25, DPR, knowledge graphs, AST indexing, NDCG / MRR / Recall@K / Acc@k / ROUGE / BERTScore

Backend & Cloud: FastAPI, REST APIs, PostgreSQL, Supabase, Next.js, Docker, CI/CD, AWS (Lambda, S3), Git, pytest, XGBoost, PyTorch

Education

University of Massachusetts Amherst — M.S. Computer Science, GPA 3.83/4.00

Sep 2024 - May 2026

Coursework: Machine Learning, Natural Language Processing (NLP), Information Retrieval, Artificial Intelligence

National Tsing Hua University — M.S. Physics

Sep 2020 - Jan 2023

National Dong Hwa University — B.S. Electrical Engineering, Minor in Computer Science

Sep 2016 - Jun 2020